| Problem | Penalty | Free | Informed | COLSHADE | L-SHADE Deb | L-SHADE pen. | CMA-ES Deb | CMA-ES pen. | Random |
|---|---|---|---|---|---|---|---|---|---|
| lens/original | 0.611 ± 0.015 | **0.692 ± 0.008** | 0.641 ± 0.037 | 0.582 | 0.586 | 0.585 | 0.562 | 0.556 | 0.555 |
| lens/realistic_draft | 0.662 ± 0.023 | **0.675 ± 0.021** | 0.647 ± 0.013 | 0.579 | 0.582 | 0.587 | 0.565 | 0.555 | 0.551 |
| cec/RC12 | 0.516 ± 0.005 | 0.718 ± 0.049 | **0.732 ± 0.056** | 0.643 | 0.621 | 0.512 | 0.505 | 0.490 | 0.372 |
| cec/RC13 | **0.963 ± 0.017** | 0.955 ± 0.005 | 0.953 ± 0.007 | 0.849 | 0.795 | 0.799 | 0.655 | 0.661 | 0.605 |
| cec/RC15 | 0.940 ± 0.031 | **0.960 ± 0.010** | 0.953 ± 0.006 | 0.711 | 0.696 | 0.699 | 0.603 | 0.605 | 0.539 |
| cec/RC16 | **0.563 ± 0.036** | 0.561 ± 0.055 | 0.516 ± 0.023 | 0.486 | 0.496 | 0.494 | 0.454 | 0.423 | 0.418 |
| cec/RC18 | **0.648 ± 0.041** | 0.622 ± 0.015 | 0.635 ± 0.033 | 0.591 | 0.606 | 0.645 | 0.515 | 0.539 | 0.506 |
| cec/RC21 | **0.935 ± 0.036** | 0.885 ± 0.049 | 0.926 ± 0.018 | 0.811 | 0.743 | 0.738 | 0.611 | 0.612 | 0.540 |
| cec/RC25 | 0.511 ± 0.004 | **0.535 ± 0.014** | 0.529 ± 0.009 | 0.515 | 0.514 | 0.506 | 0.497 | 0.496 | 0.494 |
| cec/RC26 | **0.557 ± 0.017** | 0.518 ± 0.012 | 0.513 ± 0.009 | 0.375 | 0.427 | 0.446 | 0.505 | 0.547 | 0.144 |
| Mean | 0.691 ± 0.008 | **0.712 ± 0.003** | 0.705 ± 0.008 | 0.614 | 0.607 | 0.601 | 0.547 | 0.548 | 0.472 |

*Test AOCC (higher is better). LLaMEA: mean ± std over 5 runs; baselines: mean over 15 seeds.*

*Bold: best in row.*